MICROBIOLOGY · MCQ SET

Retroviruses & HIV/AIDS

SGL 3 · Microbiology/Physiology/Genetics Dept · GUE BLOCK · PRACTICE MCQs

20 questions, mixed difficulty, 8 case-based. Attempt every question before checking the answer key on the last page.

4	10	6
Easy	Medium	Hard

1. EASY

What does the prefix 'retro' in retrovirus refer to?

A. The virus's spherical shape
B. The backward transfer of information from RNA to DNA
C. Its ability to infect backward through a cell's cytoplasm
D. Retrograde axonal transport
E. Its discovery decades after other viruses

2. EASY

Which genus and family does HIV belong to?

A. Genus Lentivirus, family Retroviridae
B. Genus Oncovirus, family Retroviridae
C. Genus Spumavirus, family Retroviridae
D. Genus Lentivirus, family Flaviviridae
E. Genus Oncovirus, family Flaviviridae

3. EASY

Which cell surface receptor is required, together with a co-receptor, for HIV entry?

A. CD8
B. CD4
C. CD19
D. CD3
E. CD16

4. EASY

Which test is the gold-standard confirmatory test for HIV infection?

A. ELISA
B. Rapid oral test
C. Western blot
D. CD4:CD8 ratio
E. Complete blood count

5. MEDIUM

Which two co-receptors does HIV use alongside CD4, and which is preferentially used in early versus late disease?

A. CCR5 (early), CXCR4 (late)
B. CXCR4 (early), CCR5 (late)
C. CCR5 only, used throughout disease
D. CXCR4 only, used throughout disease
E. CCR2 (early), CCR5 (late)

6. MEDIUM CASE

A patient with known HIV infection has a CD4 count of 180 cells/µl.

What is the most appropriate interpretation of this isolated finding?

A. He must have active tuberculosis
B. He is at increased risk of AIDS-defining illness, opportunistic infection, or malignancy
C. His CD4 count is normal and reassuring
D. This finding indicates recent, not chronic, infection
E. This value excludes HIV infection

7. MEDIUM

A patient has a positive screening test. What is the correct next step according to standard diagnostic practice?

A. Report the result as diagnostic without further testing
B. Repeat the same ELISA once more and stop if negative
C. A positive or indeterminate screening test should proceed to a confirmatory test (Western blot and/or RT-PCR)
D. Proceed directly to antiretroviral therapy without confirmation
E. Request only a CD4 count to confirm diagnosis

8. MEDIUM

Which statement about HIV transmission is CORRECT?

A. HIV can be transmitted by casual contact such as hugging or sharing food

B. HIV can be transmitted by sharing contaminated needles among IV drug users

C. HIV is transmitted through insect bites

D. HIV is transmitted through swimming pools

E. HIV is transmitted by coughing or sneezing

9. MEDIUM CASE

A patient had a high-risk needlestick exposure 10 days ago. An HIV antibody ELISA performed today is negative.

What is the most appropriate recommendation?

A. No further testing is needed, since the ELISA is negative

B. Repeat antibody testing at 6 weeks, 12 weeks and 6 months, per CDC follow-up recommendations

C. Diagnose him as HIV-negative permanently

D. Start antiretroviral therapy immediately regardless of the result

E. Order a CD4:CD8 ratio instead of repeat serology

10. MEDIUM

Which viral enzyme is responsible for inserting HIV dsDNA into the host chromosomal DNA, forming the provirus?

A. Reverse transcriptase

B. Protease

C. Integrase

D. RNA polymerase

E. DNA ligase

11. MEDIUM

Which viral enzyme cleaves HIV polyproteins during the final budding/release step to yield a mature, infectious virion?

A. Reverse transcriptase

B. Integrase

C. Viral protease

D. Helicase

E. Topoisomerase

12. MEDIUM CASE

A patient was infected with HIV approximately 2 weeks ago.

Which test result pattern would be expected at this time point, and why?

A. Antibody positive, antigen negative, because seroconversion is immediate

B. Antibody negative, p24 antigen/HIV-RNA positive, because this is within the window period with high viremia

C. Both antibody and antigen negative, because infection takes months to become detectable at all

D. Western blot positive before any ELISA is performed

E. CD4 count below 200 immediately after exposure

13. HARD

Which three mechanisms does HIV use to destroy CD4+ cells, according to the lecture?

A. Direct lysis by complement alone

B. Viral replication damaging the cell membrane, syncytium formation from cell fusion, and antibody/complement-mediated destruction of infected cells

C. Apoptosis induced solely by cytotoxic T cells

D. Phagocytosis by neutrophils alone

E. Direct destruction by antifungal antibodies

14. HARD CASE

A patient seroconverts and begins forming antibodies against major HIV antigens over the following weeks.

Which immunoglobulin specificities would be expected to appear FIRST in this patient's antibody response?

A. Antibodies to gp41 and p120

B. Antibodies to the gag proteins p24 and p55

C. Antibodies to CCR5 and CXCR4

D. Antibodies to reverse transcriptase only

E. No antibodies form until late-stage AIDS

15. HARD

Which combination of viral antigens, if reactive, would be examined on a Western blot to help confirm HIV infection?

A. gp160, gp120, p66, p55, p51, gp41, p31, p24, p17, p15

B. Only p24 and gp41

C. Only CD4 and CCR5

D. Hemagglutinin and neuraminidase

E. Only reverse transcriptase and integrase

16. HARD CASE

A patient with HIV, previously in a stable asymptomatic chronic phase for 10 years, now develops opportunistic infections and worsening symptoms.

What finding on repeat laboratory testing would best explain his clinical deterioration?

A. Rising CD4 count and falling viral load

B. Rising viral load and falling CD4 count, consistent with progression from asymptomatic chronic infection to AIDS

C. Normal, stable CD4 and viral load with no change

D. Isolated low CD8 count with normal CD4

E. A negative HIV antibody test, excluding HIV as the cause

17. HARD

Which laboratory marker correlates with response to antiretroviral therapy and is most helpful specifically before seroconversion?

A. CD4:CD8 ratio

B. Viral load (HIV-RNA) testing

C. Western blot band pattern

D. ELISA antibody titer

E. Complete blood count

18. HARD CASE

A patient has a baseline HIV-1 RNA viral load of 40,000 copies/ml.

Based on baseline viral load risk data described in the lecture, what can be inferred about his risk of progression to AIDS within 5 years compared to a patient with a baseline viral load under 4530 copies/ml?

A. His risk is lower

B. His risk is similar

C. His risk is higher, since higher baseline viral load correlates with a greater percentage progressing to AIDS within 5 years

D. Viral load has no relationship to progression risk

E. Progression risk can only be determined by CD4 count, not viral load

19. HARD

A patient's CD4 percentage drops below 20%, with the CD4:CD8 ratio reversing from its normal pattern. What was the normal ratio, and what does it become?

A. Normal 1:2, reversing to 2:1

B. Normal 2:1, reversing to 1:2

C. Normal 1:1, unchanged

D. Normal 3:1, reversing to 1:3

E. The ratio is not clinically meaningful

20. HARD CASE

A patient with advanced, progressive HIV disease is found to have declining HIV antibody levels despite ongoing infection.

What is the most likely explanation for this discrepancy?

A. B cells have decreased response to antigens, possibly due to viral proteins blocking T-cell/B-cell interaction at the CD4 site, as disease progresses

B. The patient has developed a new, unrelated autoimmune disease

C. Antibody levels always increase steadily throughout untreated HIV infection

D. This pattern indicates the patient was never truly infected

E. CD8 cells are solely responsible for antibody production, and their decline explains this

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	B	'Retro' refers to the backward transfer of genetic information from RNA to DNA via reverse transcriptase.
2	A	HIV belongs to genus Lentivirus, family Retroviridae.
3	B	CD4 is the primary receptor required, together with a co-receptor (CCR5 or CXCR4), for HIV entry.
4	C	Western blot is the gold-standard confirmatory test for HIV, with specificity >99.9% when combined with ELISA.
5	A	CCR5 is preferentially used in early disease and CXCR4 in late disease, alongside CD4.
6	B	A CD4 count below 200/µl indicates severe immune problems with high risk of AIDS-defining illness, opportunistic infection, or malignancy.
7	C	A positive or indeterminate screening test should be followed by a confirmatory test (Western blot and/or RT-PCR) per the testing algorithm.
8	B	Sharing contaminated needles is a recognized transmission route; casual contact, insect bites, swimming pools, and respiratory droplets are explicitly stated as NOT transmitting HIV.
9	B	A negative test within the window period does not exclude infection; CDC-recommended follow-up testing at 6 weeks, 12 weeks, and 6 months is appropriate.
10	C	Integrase inserts HIV dsDNA into host chromosomal DNA, forming the provirus.
11	C	Viral protease cleaves polyproteins during budding/release to yield a mature, infectious virion.
12	B	At 2 weeks post-infection, the patient is likely within the window period — antibody negative but with high viremia/ antigenemia detectable by p24 antigen and HIV-RNA assays.
13	B	HIV destroys CD4+ cells via membrane damage from viral replication, syncytium formation from cell fusion, and antibody/ complement-mediated destruction of infected cells.
14	B	Antibodies to the gag proteins p24 and p55 are the first to be detected, followed by antibodies to p51, p120 and gp41.
15	A	Western blot separates and examines reactivity to gp160, gp120, p66, p55, p51, gp41, p31, p24, p17 and p15.
16	B	Progression from asymptomatic chronic infection to AIDS is characterized by rising viral load and falling CD4 count, which would explain new opportunistic infections and clinical deterioration.
17	B	Viral load (HIV-RNA) testing is most helpful before seroconversion and correlates with response to antiretroviral therapy.
18	C	Higher baseline viral load correlates with a higher percentage of patients progressing to AIDS within 5 years, so a viral load of 40,000 copies/ml confers higher risk than one under 4530 copies/ml.
19	B	The normal CD4:CD8 ratio is 2:1, reversing to 1:2 when CD4 percentage drops below 20%.
20	A	As HIV disease progresses, antibody levels decrease; B cells show decreased antigen response, possibly from viral proteins blocking T-cell/B-cell interaction at the CD4 site.